



भारतीय प्रौद्योगिकी संस्थान पटना
Indian Institute of Technology Patna

EDUNEXA: DESIGN AND DEVELOPMENT OF A SMART PLATFORM FOR SKILL-BASED ONLINE TUTORING

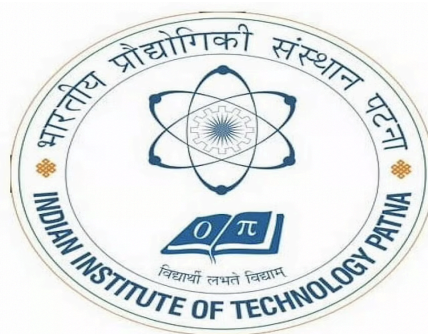
Capstone-I Project Report Submitted
by the student of

Hybrid UG Program in Computer Science & Data Analytics

INDIAN INSTITUTE OF TECHNOLOGY PATNA

Bihta - 801106, India

Date: 25/03/2026





1. Declaration

I hereby declare that this submission is my own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

Date: 25/03/2026

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Group No: 66

Signature:

2. Summary of the Project

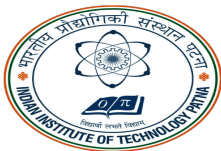
EduNexa is a smart skill-based online tutoring platform designed to connect students with qualified teachers in an efficient and structured manner. The platform consists of two primary modules: a teacher module and a student module.

In the teacher module, educators can register themselves, provide their qualifications, skills, and expertise, and create their professional profiles. In the student module, learners can register, access personalized dashboards, track their academic progress, and enroll in courses based on their interests.

The system leverages modern web technologies including frontend frameworks, backend APIs, and database systems to ensure scalability, performance, and user-friendly interaction. The main objective of this project is to create a centralized platform that enhances skill-based learning and provides real-time insights into student performance.

Work Done by Each Member

S.No	Team Member	Role / Responsibility	Enrollment No.
1	Arun Kumar Giri	Full Stack Development (Frontend, Backend, Database Design), DevOps (Deployment & Integration), and Project Lead	UA2504AIH48
2	Ajeet Kumar Gautam	Responsible for designing user interfaces and UI-UX.	ua2504aih26



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3	Shailesh Kumar Giri	bug fixing, and performance validation.	UA2503AIH405
4	Amit Kumar Devanda	Documentation and Research. Responsible,--research analysis	ua2504aih32
5	Aditya Kumar Dwivedi	Presentation – PPT	ua2504aih19



3. Introduction

In recent years, the demand for skill-based education has significantly increased due to rapid technological advancements. Traditional learning systems often lack personalization and real-time performance tracking, which limits effective learning outcomes.

EduNexa aims to address these issues by providing a digital platform where students can learn from qualified teachers while tracking their growth through an interactive dashboard. The platform focuses on:

- Connecting students with skilled instructors
- Providing structured learning paths
- Offering real-time progress monitoring

The system is designed using modern web technologies to ensure scalability and efficiency. The motivation behind EduNexa is to reduce the gap between learners who seek skill development and educators who possess domain expertise, enabling a seamless exchange of knowledge in a digital-first world.

The EduNexa platform is built to serve a wide range of users — from students pursuing academic excellence to professionals seeking domain-specific skills. By offering an intelligent matching mechanism between learners and teachers, the system promotes personalized education tailored to individual needs and goals.

4. System Overview

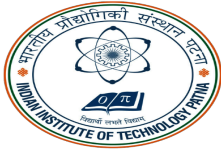
The EduNexa platform is divided into two main components:

Teacher Section

- Registration and authentication
- Profile creation with qualifications
- Course or skill listing
- Availability management

Student Section

- Registration and login
- Personalized dashboard
- Course enrollment
- Progress tracking



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The system follows a client-server architecture where the frontend communicates with backend APIs, and data is stored in a centralized database. The platform leverages React.js for the frontend, Node.js for the backend, and MongoDB as the primary database, ensuring a scalable and maintainable technology stack.

Technology Stack

Layer	Technology	Purpose
Frontend	React.js	Dynamic and responsive user interface
Backend	Node.js + Express	RESTful API development and server logic
Database	MySQL	data storage and management
Authentication	JWT Tokens	Secure session and access management
Hosting	Cloud Platform	Deployment and scalability support



5. Conclusion & Future Work

EduNexa provides a scalable and efficient platform for skill-based online tutoring. It simplifies the interaction between students and teachers while enabling personalized learning experiences. The platform successfully integrates modern technologies to deliver a smooth and responsive user experience.

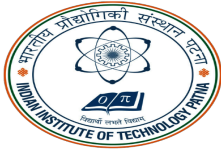
The EduNexa platform demonstrates how technology can transform education by making it more accessible and personalized. It ensures better engagement between students and teachers and provides valuable insights into learning progress. The system architecture supports horizontal scaling, making it suitable for large user bases.

Future Enhancements

- AI-based course recommendation system
- Live video classes integration
- Payment gateway for premium courses
- Mobile application development
- Multi-language support for regional learners
- Gamification features for improved student engagement

6. References

- [1] React.js Official Documentation — <https://reactjs.org/docs>
- [2] Node.js Documentation — <https://nodejs.org/en/docs>
- [3] MongoDB Documentation — <https://www.mongodb.com/docs>
- [4] Research Papers on Online Learning Platforms
- [5] IEEE Journals on Educational Technology
- [6] Express.js Documentation — <https://expressjs.com>
- [7] JWT Authentication — <https://jwt.io/introduction>



Contribution Matrix

Team Member	Frontend	Backend	Database	Deployment	Report Writing	PPT	Overall Role
Arun Kumar Giri	✓	✓	✓	✓	✓	✓	Project Lead (Full Stack Development)
Ajeet Kumar Gautam					✓	✓	UI-UX Design
Shailesh Kumar Giri					✓	✓	Bug fixing
Amit Kumar Devanda					✓	✓	Report Writing
Aditya Kumar Dwivedi					✓	✓	Presentation